

Classification: Accuracy, recall, precision, and related metrics

True and false positives and negatives are used to calculate several useful metrics for evaluating models. Which evaluation metrics are most meaningful depends on the specific model and the specific task, the cost of different misclassifications, and whether the dataset is balanced or imbalanced.

All of the metrics in this section are calculated at a single fixed threshold, and change when the threshold changes. Very often, the user tunes the threshold to optimize one of these metrics.

Accuracy is the proportion of all classifications that were correct, whether positive or negative. It is mathematically defined as:

$$\text{Accuracy} = \frac{\text{correct classifications}}{\text{total classifications}} = \frac{TP + TN}{TP + TN + FP + FN}$$

In the spam classification example, accuracy measures the *fraction of all emails correctly classified*.

A perfect model would have zero false positives and zero false negatives, and therefore an accuracy of 1.0, or 100%.

Because it incorporates all four outcomes from the [confusion matrix](#) (TP, FP, TN, FN), given a balanced dataset, with similar numbers of examples in both classes, accuracy can serve as a coarse-grained measure of model quality. For this reason, it is often the default evaluation metric used for generic or unspecified models carrying out generic or unspecified tasks.

However, when the dataset is imbalanced, or where one kind of mistake (FN or FP) is more costly than the other, which is the case in most real-world applications, it's better to optimize for one of the other metrics instead.

For heavily imbalanced datasets, where one class appears very rarely, say 1% of the time, a model that predicts negative 100% of the time would score 99% on accuracy, despite being useless.

Recall, or true positive rate

The **true positive rate (TPR)**, or the proportion of all actual positives that were classified correctly as positives, is also known as [recall](#).

Recall is mathematically defined as:

$$\text{Recall (or TPR)} = \frac{\text{correctly classified actual positives}}{\text{all actual positives}} = \frac{TP}{TP + FN}$$

False negatives are actual positives that were misclassified as negatives, which is why they appear in the denominator. In the spam classification example, recall measures the *fraction of spam emails that were correctly classified as spam*. This is why another name for recall is **probability of detection**: it answers the question "What fraction of spam emails are detected by this model?"

A hypothetical perfect model would have zero false negatives and therefore a recall (TPR) of 1.0, which is to say, a 100% detection rate.

In an imbalanced dataset where the number of actual positives is very low, recall is a more meaningful metric than accuracy because it measures the ability of the model to correctly identify all positive instances. For applications like disease prediction, correctly identifying the positive cases is crucial. A false negative typically has more serious consequences than a false positive. For a concrete example comparing recall and accuracy metrics, see the notes in the definition of [recall](#).

False positive rate

The **false positive rate (FPR)** is the proportion of all actual negatives that were classified *incorrectly* as positives, also known as the **probability of false alarm**. It is mathematically defined as:

$$\text{FPR} = \frac{\text{incorrectly classified actual negatives}}{\text{all actual negatives}} = \frac{FP}{FP + TN}$$

False positives are actual negatives that were misclassified, which is why they appear in the denominator. In the spam classification example, FPR measures the *fraction of legitimate emails that were incorrectly classified as spam*, or the model's rate of false alarms.

A perfect model would have zero false positives and therefore a FPR of 0.0, which is to say, a 0% false alarm rate.

For an imbalanced dataset, FPR is generally a more informative metric than accuracy. However, if the number of actual negatives is very low, FPR may not be an ideal choice, due to its volatility. For example, if there are only four actual negatives in a dataset, one misclassification results in an FPR of 25%, while a second misclassification causes the FPR to jump to 50%. In cases like this, [precision](#) (described next) may be a more stable metric for evaluating the effects of false positives.

Precision

Precision is the proportion of all the model's positive classifications that are actually positive. It is mathematically defined as:

$$\text{Precision} = \frac{\text{correctly classified actual positives}}{\text{everything classified as positive}} = \frac{TP}{TP + FP}$$

In the spam classification example, precision measures the *fraction of emails classified as spam that were actually spam*.

A hypothetical perfect model would have zero false positives and therefore a precision of 1.0.

In an imbalanced dataset where the number of actual positives is very, very low, say 1-2 examples in total, precision is less meaningful and less useful as a metric.

Precision improves as false positives decrease, while recall improves when false negatives decrease. But as seen in the previous section, increasing the classification threshold tends to decrease the number of false positives and increase the number of false negatives, while decreasing the threshold has the opposite effects. As a result, precision and recall often show an inverse relationship, where improving one of them worsens the other.

Try it yourself:

What does NaN mean in the metrics?

NaN, or "not a number," appears when dividing by 0, which can happen with any of these metrics. When TP and FP are both 0, for example, the formula for precision has 0 in the denominator, resulting in NaN. While in some cases NaN can indicate perfect performance and could be replaced by a score of 1.0, it can also come from a model that is practically useless. A model that never predicts positive, for example, would have 0 TPs and 0 FPs and thus a calculation of its precision would result in NaN.

Choice of metric and tradeoffs

The metric(s) you choose to prioritize when evaluating the model and choosing a threshold depend on the costs, benefits, and risks of the specific problem. In the spam classification example, it often makes sense to prioritize recall, nabbing all the spam emails, or precision, trying to ensure that spam-labeled emails are in fact spam, or some balance of the two, above some minimum accuracy level.

Metric	Guidance
Accuracy	Use as a rough indicator of model training progress/convergence for balanced datasets. For model performance, use only in combination with other metrics. Avoid for imbalanced datasets. Consider using another metric.
Recall (True positive rate)	Use when false negatives are more expensive than false positives.
False positive rate	Use when false positives are more expensive than false negatives.
Precision	Use when it's very important for positive predictions to be accurate.

(Optional, advanced) F1 score

The **F1 score** is the harmonic mean (a kind of average) of precision and recall.

Mathematically, it is given by:

$$F1 = 2 * \frac{\text{precision} * \text{recall}}{\text{precision} + \text{recall}} = \frac{2TP}{2TP + FP + FN}$$

This metric balances the importance of precision and recall, and is preferable to accuracy for class-imbalanced datasets. When precision and recall both have perfect scores of 1.0, F1 will also have a perfect score of 1.0. More broadly, when precision and recall are close in value, F1 will be close to their value. When precision and recall are far apart, F1 will be similar to whichever metric is worse.

Exercise: Check your understanding

A model outputs 5 TP, 6 TN, 3 FP, and 2 FN. Calculate the recall.

0.455

0.714

0.625

A model outputs 3 TP, 4 TN, 2 FP, and 1 FN. Calculate the precision.

0.75

0.6

0.429

You're building a binary classifier that checks photos of insect traps for whether a dangerous invasive species is present. If the model detects the species, the entomologist (insect scientist) on duty is notified. Early detection of this insect is critical to preventing an infestation. A false alarm (false positive) is easy to handle: the entomologist sees that the photo was misclassified and marks it as such. Assuming an acceptable accuracy level, which metric should this model be optimized for?

Precision

Recall

False positive rate (FPR)